

Microstrip Patch Antenna Design with Improved gain using Geometric Optimization

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Abstract—This paper presents a novel cut microstrip patch antenna (MSP) for enhanced gain. The design suggests a special defected ground structure (DGS) in the microstrip patch antenna for a wide frequency range. The design is tested using CST microwave studio and measured at a frequency of 4 GHz. The improved antenna operates in the frequency range of 3.8 GHz to 4.22 GHz providing 11% impedance bandwidth. The paper discusses the evolution of the final prototype and the simulated and measured results agree. The final prototype provides a high gain of 3.3 dBi for a simple yet effective design.

Index Terms—Patch antenna, defected ground structure, high gain, wideband

I. INTRODUCTION

The demand for compact, lightweight, and efficient antennas has led to the evolution of the wireless communication system into its present magnificence. Figure 1 shows the prototype of the proposed antenna. Microstrip patch antennas (MSP) have been the new appearances among the candidates for these developments due to their being produced in the low profile, ease of fabrication, being substituted for the integrated circuits, etc. Initially invented in 1971 for microwave circuits, MSP antennas have gone through an evolutionary process to be antennas that nowadays are fully developed and used in various applications, such as satellite communication, mobile phones, radar systems, and biomedical devices. This paper aims to analyze the design and modeling of MSP antennas emphasizing gain and radiation frequency improvement.

To improve the performance of MSP antennas, the use of design techniques such as Defected Ground Structure (DGS) and tapered patches is envisaged. The technique DGS, on the other hand, means it involves

the introduction of intentional defects or discontinuities in the ground plane, which then alters the propagation of electromagnetic waves thus leading to an enhanced antenna performance in terms of bandwidth, radiation patterns, and efficiency [10]. The tapered patch technique, on the other hand, implies the modification of the patch shape to achieve the specified radiation characteristics and frequency response [11]. First, we will present a

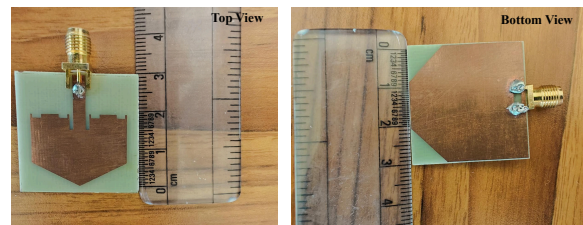


Fig. 1. Prototype of proposed antenna

detailed description of MSP antennas, their mode of operation, as well as their advantages and limitations. Then, different design techniques and methodologies are reported, starting from the simple single-layer patches to the advanced tapered patches [8]. The procedure for selecting the most suitable design considering the frequency and application requirements is displayed in detail. The suggested design in the proposed research is done through the placement of a defected ground plane (DGS) in the microstrip patch antenna which is used to extend the operational bandwidth and, specific to radiation pattern, see an overall efficacy. The DGS introduces distortion in the electromagnetic wave distribution path, the result being an improved antenna. Defective ground

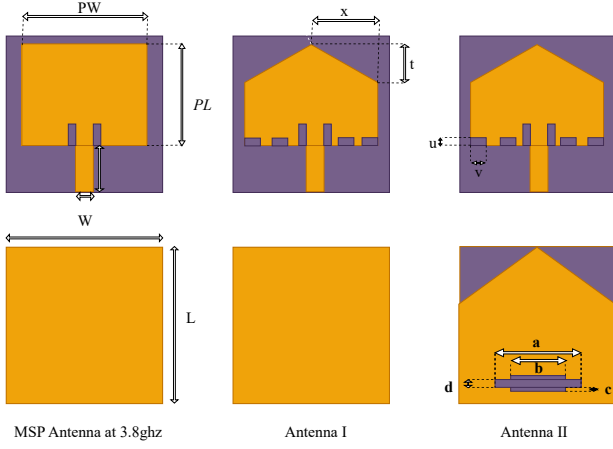


Fig. 2. Evolution of Microstrip patch antenna

planes make the frequency range wider, thus eliminating the issues with the conventional Microstrip antennas that were typically confined to narrow bandwidths [7].

II. PROPOSED CONFIGURATION

1) *Tapered rectangular patch antenna*: The shape of a trapezoid is formed in this case by reducing the patch along its edges. The amplitude of tapering will help us to control the necessary frequency and gain for it. An example of such a design is shown in Figure 2.

Simulations are carried out using software package i.e CST Microwave studio to get optimum values for the dimensions of the patch length and tapering patch cuts. In order to obtain maximum possible gain with minimum return loss, we should tune these parameters as well as consider matching conditions.

Here are the dimensions of the proposed tapered patch antenna in table I:

TABLE I
PROPOSED ANTENNA DIMENSIONS (MM)

Parameter	Value
Patch Length (L)	22
Patch Width (W)	18
Tapering Angle (θ)	45°
Substrate Height (h)	1.6
Substrate Dielectric Constant (ϵ_r)	4.4
Feed Line Width (W_f)	3
Feed Line Length (L_f)	10

These dimensions are for the proposed antenna and can be adjusted based on the desired frequency and gain characteristics. The tapering of patch can be adjusted to control the gain of the antenna, while the patch dimensions and substrate properties can be adjusted to control the resonant frequency. In figure 3 the parametric analysis on the proposed antenna shows that the required results were found to be at $x=6.06\text{mm}$ and $t=12.115\text{mm}$.

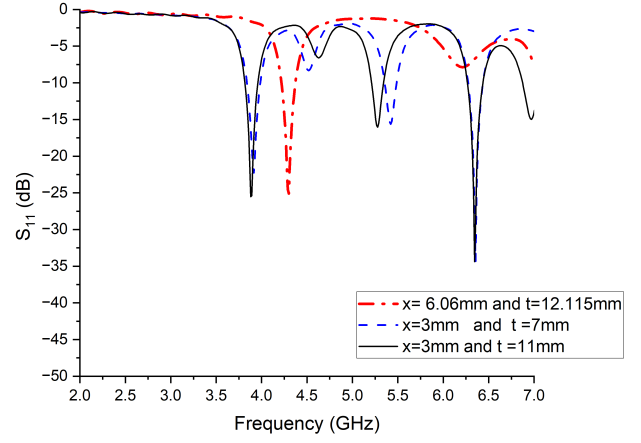


Fig. 3. Parametric analysis for S-parameter on changes on patch cuts x, t

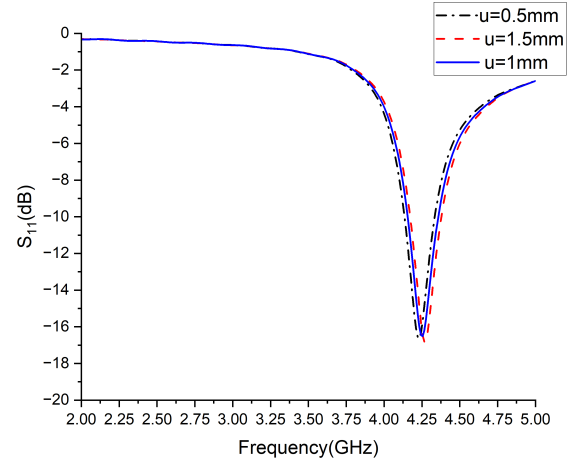


Fig. 4. Parametric analysis of edge cuts for value of u

Fig 5 represents the measured and simulated comparison of return loss S_{11} for antenna I and II i.e the proposed antenna. har Compared with a basic rectangular patch design, using a tapered patch design can enhance the gain and frequency response of the Microstrip patch antenna. In figure 4 the parametric analysis on the edge cuts for value of u on proposed antenna shows for $u=1\text{mm}$. A more intricate current distribution is created on the patch surface because the patch edges are tapered. In Figure 6 the Electric field is checked for the proposed antenna and the patch is tapered where surface current is minimum. This results in higher gain, better radiation characteristics.

2) *Defected ground structure*: The DGS can be implemented in various ways, such as by etching slots or cuts in the ground plane, or by adding patches or other structures to the ground plane. In the proposed antenna,

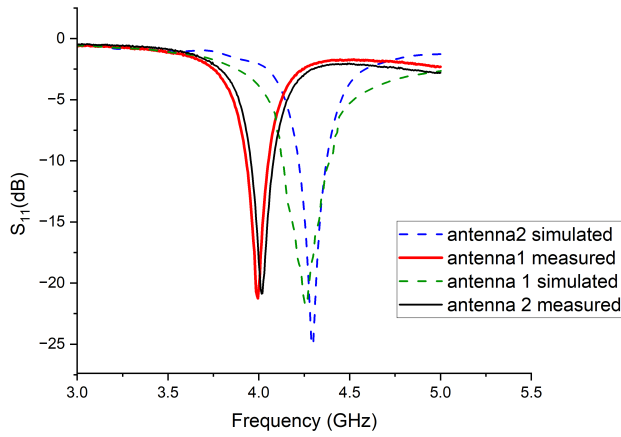


Fig. 5. Measured and Simulated comparison of return loss S_{11} for antenna I and antenna II

using a DGS while cutting rectangular holes named notch cuts beside the inset feed line is into the ground plane. Simulations using CST microwave studio can be used to design the cuts of necessary sizes and at the right locations by analysing through parametric analysis. The triangular cuts are located more close to the upper edges of the ground plane and other types of cuts are located near the feed point of the antenna, the qualities of the sliced grounds are targeted, of the designated performance characteristics.

TABLE II
COMPARATIVE STUDY WITH DIFFERENT PUBLICATIONS

Reference	Patch Type	Operating Frequency (GHz)	DGS	Gain (dBi)	ϵ_r
[7]	Rectangular	5.9	Yes	2.5	4.9
[10]	Rectangular	3.5	yes	2.6	4.4
[11]	Rectangular	3.45	no	3.83	4.3
Proposed Antenna	Rectangular	3.8	yes	4.3	4.4

The table II shows the study of different publications and by that analyzing various characteristics through the study.

DGS are credited with enhancement of the antenna by the fact that, for example, it produces the antenna's bandwidth by decreasing the antenna's quality factor. The DGS also increase the antenna's radiation patterns, thus limiting the stray surface waves that can be imprisoned within the substrate, and it also reduces the DGS's gain by the improvement of the surprising gradient which is further associated with the antenna and the feed line through the impedance matching technique.

Table III and Figure 9 shows the parametric analysis for the DGS notch cut and for the value for $a=8\text{mm}$, $b=6\text{mm}$, $c=0.5\text{mm}$, $d=1\text{mm}$ the results are optimised.

In the case of your proposed antenna, you can consider implementing a DGS by introducing rectangular cuts or

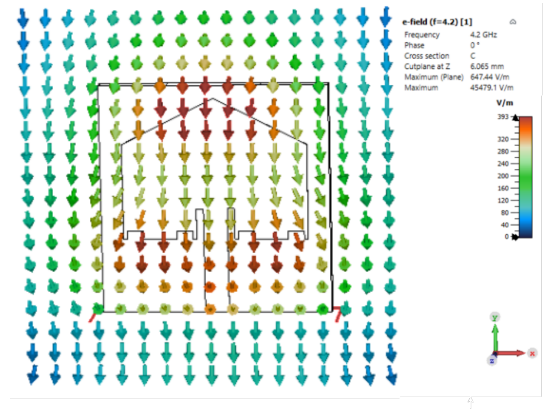


Fig. 6. E-field for proposed antenna

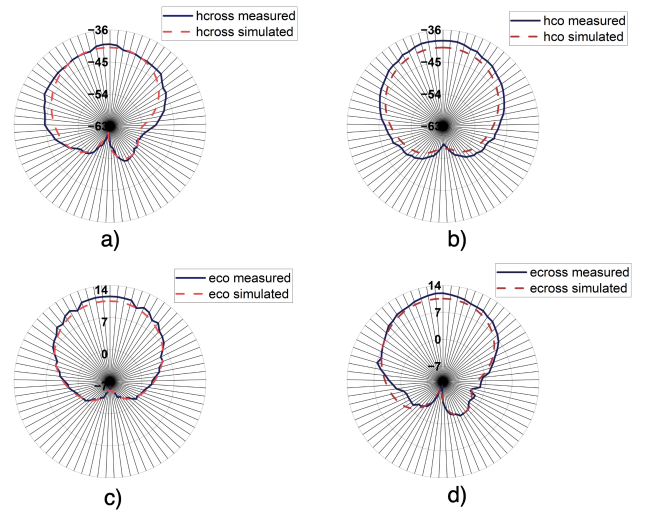


Fig. 7. Radiation pattern of antenna 1, a)hcross, b)hco, c)eco, d)ecross

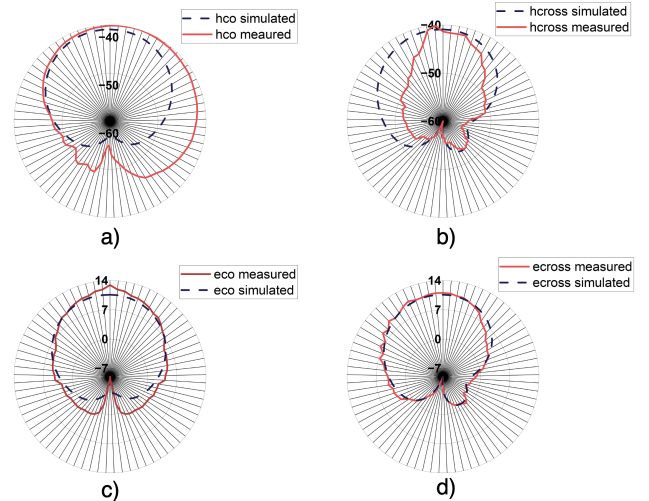


Fig. 8. Radiation pattern of proposed antenna, a)hco, b)hcross, c)eco, d)ecross

TABLE III
PARAMETRIC ANALYSIS FOR DGS NOTCH CUT

Parameter	Original Value (mm)	1st (mm)	2nd (mm)
a	8	11	7
b	6	7	5
c	0.5	0.5	1
d	1	1	1

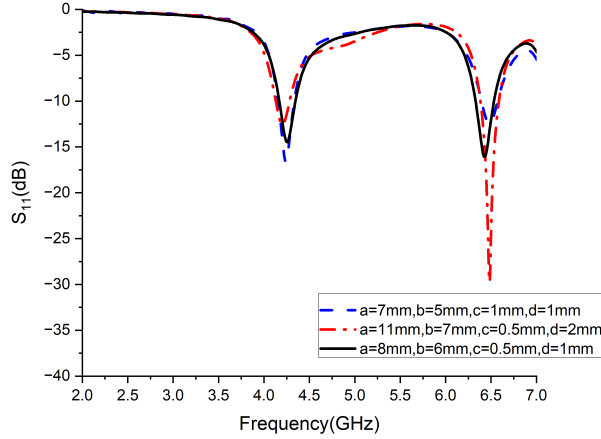


Fig. 9. Parametric analysis of DGS notch cut

slots in the ground plane and optimizing their size and location using simulation software.

Fig 7 and 8 depicts the radiation pattern that has been measured and simulated for the antenna1 and the proposed antenna. For which the proposed antenna has 6.24 GHz of main lobe gain. Figure 10 depicts the comparison in gain for antenna 1 and the proposed antenna in the case of the proposed antenna, we considered implementing DGS by introducing rectangular cuts or

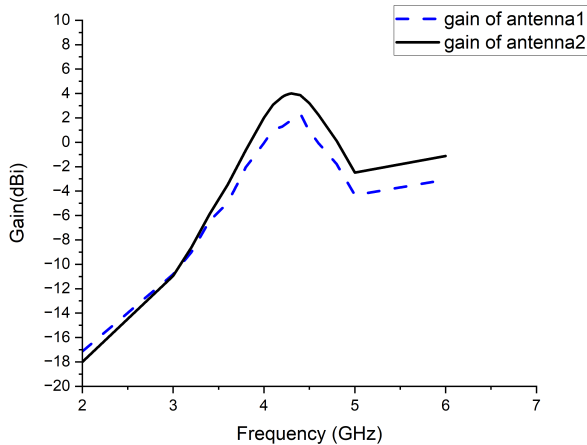


Fig. 10. Comparison of gain for antenna 1 and proposed antenna 2

slots in the ground plane and optimizing their size and location using simulation software.

III. CONCLUSION

The simple patch antenna, at an operating frequency of 3.8 GHz, and the proposed antenna has 4.27 GHz which has gain increased to 4.3dBi using DGS and a tapered patch. This percentage change in dB gain is 5.74 percent. In Fig. 2, the simple MSP patch antenna with a cut in the patch close to the feeding line, the cut was made to increase the capacitance at the edges due to concentration of more electric field line at edges. The increase in capacitance resulted in increasing the bandwidth of a simple patch antenna. The defective ground plane is placed below the ground plane of the antenna, it improves the gain and radiation characteristics of the antenna. These changes provide the increased gain of the antenna as well as improved bandwidth. This overall would be beneficial in various applications, such as satellite communication, mobile phones, radar systems, and biomedical devices.

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